PVsyst - Simulation report

Standalone system

Project: Nuevo Proyecto

Variant: Nueva variante de simulación

Standalone system with batteries

System power: 1160 Wp

Centro - Italy

Author

PVsyst V8.0.18
VC0, Simulation date:
06/11/25 14:29
with V8.0.18

Project summary

Geographical Site
Centro
Italy

Situation

Latitude	42.84 °(N)
Longitude	12.24 °(E)
Altitude	567 m
Time zone	UTC+1

Project settings

Albedo	0.20

Weather data
Centro
Meteonorm 8.2 (1991-2002), Sat=100% - Sintético

System summary

Standalone system

Standalone system with batteries

Orientation #1
Fixed plane
Tilt/Azimuth 20 / 0 °

User's needs
Daily household consumers
Constant over the year
Average 1.8 kWh/Day

System information

PV Array

Nb. of modules	2 units
Pnom total	1160 Wp

Battery pack

Technology	Lithium-ion, NCA
Nb. of units	1 unit
Voltage	50 V
Capacity	134 Ah

Results summary

Useful energy from solar	654.94 kWh/year	Specific production	565 kWh/kWp/year	Perf. Ratio PR	33.71 %
Missing Energy	6.23 kWh/year	Available solar energy	1656.36 kWh/year	Solar Fraction SF	99.04 %
Excess (unused)	961.46 kWh/year				

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General parameters

Standalone system — **Standalone system with batteries**

Orientation #1
Fixed plane
Tilt/Azimuth 20 / 0 °

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

User's needs
Daily household consumers
Constant over the year
Average 1.8 kWh/Day

PV Array Characteristics

PV module

Manufacturer	Generic
Model	LR5-78HBD-580M Bifacial
(Original PVsyst database)	
Unit Nom. Power	580 Wp
Number of PV modules	2 units
Nominal (STC)	1160 Wp
Modules	1 strings x 2 In series
At operating cond. (50°C)	
Pmpp	1063 Wp
U mpp	81 V
I mpp	13 A

Total PV power

Nominal (STC)	1.16 kWp
Total	2 modules
Module area	5.6 m²
Cell area	5.2 m²

Controller

Universal controller	
Technology	MPPT converter
Temp coeff.	-5.0 mV/°C/Elem.
Converter	
Maxi and EURO efficiencies	97.0 / 95.0 %

Battery Management control

Threshold commands as	SOC calculation
Charging	SOC = 0.96 / 0.80
Discharging	SOC = 0.10 / 0.35

Battery Storage

Battery

Manufacturer	Generic
Model	Powerwall
Technology	Lithium-ion, NCA
Nb. of units	1 Unit
Discharging min. SOC	10.0 %
Stored energy	6.1 kWh
Battery Pack Characteristics	
Voltage	50 V
Nominal Capacity	134 Ah (C10)
Temperature	Fixed 20 °C

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 102 mΩ
Loss Fraction 1.50 % at STC

Serie Diode Loss
Voltage drop 0.7 V
Loss Fraction 0.8 % at STC

Module Quality Loss
Loss Fraction -0.75 %

Module mismatch losses
Loss Fraction 0.50 % at MPP

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

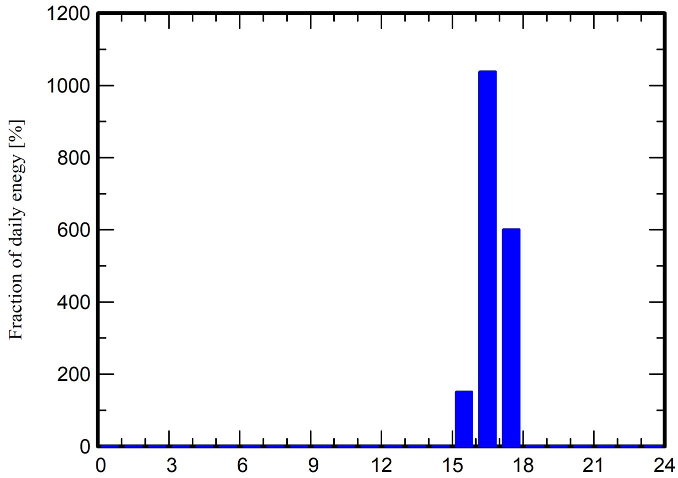
Detailed User's needs

Daily household consumers, Constant over the year, average = 1.8 kWh/day

Annual values

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lámparas (LED o fluo)	25	15/lamp	0.5	188
TV / PC / móvil	3	100/app	0.5	150
Electrodomésticos	2	250/app	0.5	250
Lavaplatos y lavadora	1		2	1200
Consumidores en espera			24.0	24
Total daily energy				1812

Hourly distribution



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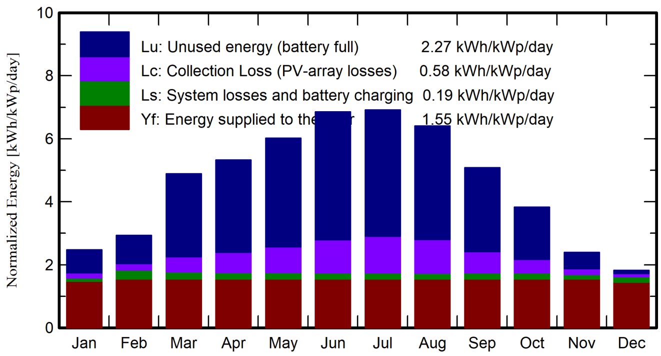
Main results

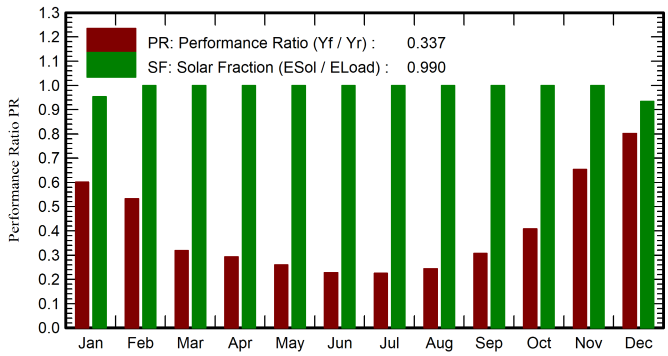
System Production

Useful energy from solar	654.94 kWh/year	Perf. Ratio PR	33.71 %
Available solar energy	1656.36 kWh/year	Solar Fraction SF	99.04 %
Excess (unused)	961.46 kWh/year		
Loss of Load		**Battery ageing (State of Wear)**	
Time Fraction	1.0 %	Cycles SOW	88.1 %
Missing Energy	6.23 kWh/year	Static SOW	90.0 %

Normalized productions (per installed kWp)



Performance Ratio PR



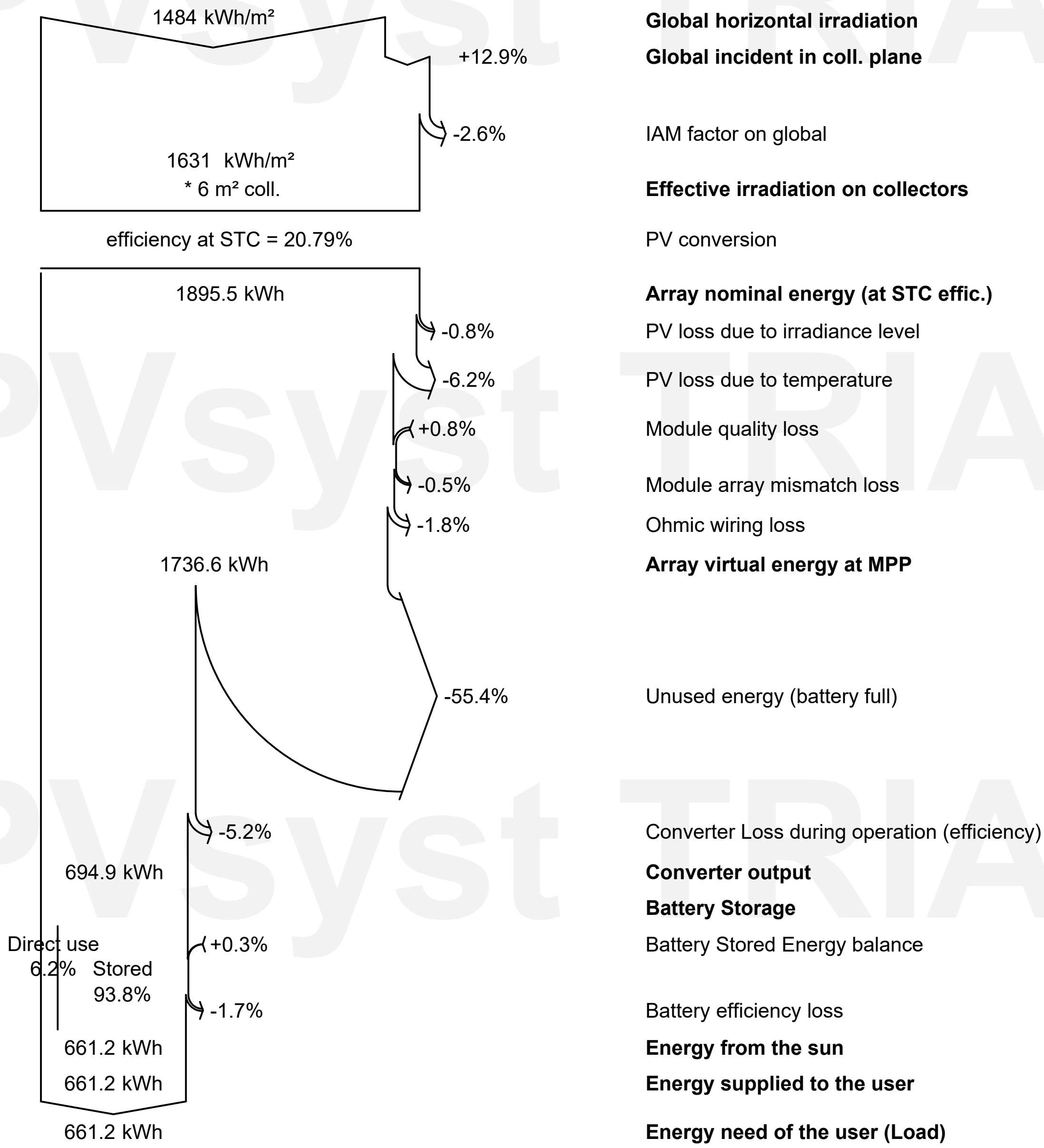
Balances and main results

	GlobHor kWh/m²	GlobEff kWh/m²	E_Avail kWh	EUnused kWh	E_Miss kWh	E_User kWh	E_Load kWh	SolFrac ratio
January	49.8	74.6	80.2	26.2	2.60	53.6	56.2	0.953
February	62.0	79.9	85.4	28.6	0.00	50.7	50.7	1.000
March	123.5	147.7	155.3	94.3	0.00	56.2	56.2	1.000
April	145.6	155.8	159.8	101.8	0.00	54.3	54.3	1.000
May	183.0	182.0	183.5	123.8	0.00	56.2	56.2	1.000
June	205.9	200.3	198.5	140.8	0.00	54.3	54.3	1.000
July	212.7	208.8	203.5	144.0	0.00	56.2	56.2	1.000
August	185.2	193.6	188.2	129.0	0.00	56.1	56.1	1.000
September	132.0	148.5	150.6	92.7	0.00	54.3	54.3	1.000
October	92.5	115.8	119.2	59.3	0.00	56.2	56.2	1.000
November	52.1	69.6	73.7	17.6	0.00	54.3	54.3	1.000
December	39.4	54.6	58.4	3.4	3.63	52.5	56.2	0.935
Year	1483.7	1631.2	1656.4	961.5	6.23	654.9	661.2	0.990

Legends

GlobHor	Global horizontal irradiation	E_User	Energy supplied to the user
GlobEff	Effective Global, corr. for IAM and shadings	E_Load	Energy need of the user (Load)
E_Avail	Available Solar Energy	SolFrac	Solar fraction (EUsed / ELoad)
EUnused	Unused energy (battery full)		
E_Miss	Missing energy		

Loss diagram



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Predef. graphs

Daily Input/Output diagram